Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/22/2005
PI Name:	Sutton, Jeffrey M.D., Ph.D.		
Project Title:	Near Infrared Brain Imaging for Space Medicine		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02129	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	06/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Boas, David (Massachusetts General Hospital) Koroshetz, Walter (Massachusetts General Hospital) Rosen, Bruce (Massachusetts General Hospital) Strangman, Gary (Massachusetts General Hospital)		
Grant/Contract No.:	NCC 9-58-SMS00202		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

This project is part of the NSBRI Smart Medical Systems Team. It develops and applies a new technology, diffuse optical tomography (DOT; also called near-infrared spectroscopy and imaging, NIRS/NIRI), for portable non-invasive functional monitoring suitable for neuroimaging in space. The technology has potential capabilities to:
• Quantitatively assess physiological adaptation (e.g., changes in intracranial pressure and blood flow) associated with microgravity; • Detect regional brain activity correlated with performance under altered circadian and mental loads; • Provide remote clinical assessment; and • Guide treatment.

As assessed by NASA, the project addresses two of the four highest priority risk areas on the Critical Path Roadmap (CPR). These are (1) trauma and acute medical problems, and (2) human performance failure because of poor psychosocial adaptation. In addition to applications for space, the developments of this project have relevance to Earth medical research and care. For example, DOT is now being tested for monitoring stroke progression in patients with acute cerebrovascular accidents.

The project brings together scientists, engineers and physicians at the Massachusetts General Hospital/Harvard-MIT Division Health Sciences and Technology, and medical operations personnel at NASA Johnson Space Center (JSC), to work collaboratively on the development and testing of DOT as a space relevant technology. The original aims are to:

1. Refine current DOT technology to build an instrument with improved spatial and temporal resolution to detect brain activity non-invasively, and in real time, through the intact skull;

2. Validate the improved instrument using functional magnetic resonance imaging (fMRI), which is a standard technology, and test DOT as a portable brain imaging device for assessing motor and cognitive activity under normal and sleep deprived conditions in normal human subjects;

3. Assess DOT, along with optical coherence tomography (OCT), to non-invasively measure altered intracranial pressure (ICP) in neurological patients, given that altered ICP may occur in the space environment; and

4. Refine a system for automated image interpretation using individualized brain models and computational techniques.

During the third and final year of this award, including supplementary extension, key findings and accomplishments include:

• Continued refinement of DOT instrumentation to validate the technology using fMRI; • Simultaneous application of DOT and fMRI to detect brain activity differences during SpaceDock task performance (a custom-designed space-relevant visuomotor performance task); • Simultaneous application of DOT and fMRI to detect brain activity differences associated with sleep deprivation; • Manuscript completed for the sleep-deprivation application data (2 manuscripts under review); • Further development of a demonstration system for automated multi-sensor analysis and interpretation (Automated Intelligent Medical System; AIMS), which can use sensor information from multiple astronauts and/or the environment to make medical diagnoses; • Manuscript completed describing the AIMS computational system and characterization results; and • Leveraging of the NSBRI funding to win an R21 award to investigate in greater detail the application of DOT in patients with altered intracranial pressure.

The impact of these findings are that they address the technology requirements set forth in Aim 1, and provide validation for DOT as laid out in the objectives and hypotheses of Aim 2. The previously acquired motor data using simultaneous DOT and fMRI confirm one of the hypotheses contained in Aim 2, namely that DOT and fMRI will be able to detect changes in regional brain activity contralateral to motor movement. The DOT/fMRI findings using the SpaceDock task confirm the hypothesis that the DOT technique can detect frontal cortex differences as a function of mental load in normal and sleep-deprived subjects. The AIMS system provides a general-purpose platform for simultaneously and autonomously monitoring many sensors, and speaks to the issue of individualized, digitized human, anatomical brain models upon which time-derivative functional data are used for interpretation and display in real time (Aim 4).

In the coming year, under the new R21 funding, patient populations with elevated ICP will be investigated in considerable detail. Finally, continued development will take place on the AIMS informatics system towards more automated machine/human interface, applicable to DOT imaging as well as other, analog settings.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Drs. Boas and Koroshetz have an active research program at the Massachusetts General Hospital using DOT for non-invasive, portable functional brain monitoring in neurological patients. A study is under way to monitor stroke progression and to apply DOT as a means to monitor treatment for rehabilitation. Drs. Sutton and Strangman also have plans for DOT applications in neuroscience and studies of cognitive dysfunction—in particular, for the monitoring and evaluation of rehabilitation from stroke or brain injury. Other medical applications for DOT are being pursued at several centers across the country.
More generally, Earth-based applications of the refined DOT and AIMS technologies are numerous. Examples (not comprehensive) appear below:

* Functional brain imaging of (or even during) rehabilitation from stroke or brain injury

* Cerebral imaging or monitoring in remote settings (rural health centers, mountainside, etc.)

* Detection of tissue ischemia (esp. muscle, abdominal, brain)

* Pediatric brain imaging/evaluation (particularly for young, highly-animated patients)

* Stroke detection and monitoring

* Breast cancer detection

* Autonomous medical diagnosis in remote (on-location) settings

Task Progress:	In the third and final year of the project, we have completed all specific aims, as described in the Specific Aims section herein. In particular: Aim 1: We have continuously refined and enhanced all aspects of DOT instrumentation for use in long-term recording settings (as would be necessary for spaceflight) as well as producing instrumentation that is far more robust to motion artifacts than any other existing brain imaging technology (also critical for applications in spaceflight). This helps demonstrate the feasibility of using DOT in a spaceflight context, and achieves the goals of Aim 1. Aim 2: We demonstrated that DOT provides information about brain function that is equivalent to the information obtained by fMRI, and that the better temporal resolution and spectroscopic character of DOT provides additional information unavailable via fMRI. We also demonstrated the applicability of DOT in a spaceflight-relevant context--as subjects performed SpaceDock under conditions of varying levels of sleep deprivation--as well as the ability of DOT (and fMRI) to detect cerebral alterations associated with sleep deprivation. Together, this achieved the goals of Aim 2. Aim 3: This aim from the original proposal could not be completed due to NSBRI resource-reduction during the lifetime of the grant. However, the data from this NSBRI project was used as preliminary data to win a follow-on grant from the NIH-NIBIB to investigate intracranial pressure in greater detail than originally proposed. This will allow us to achieve considerably more than the goals for Aim 3. Aim 4: As proposed, we completed the development of a system for autonomous evaluation of multi-sensor data--a system that was designed to be even more general and widely applicable than originally proposed. Initial characterization of this system has been completed, along with a manuscript detailing the system and characterization results. We therefore achieved the goals of Aim 4.
Bibliography Type:	Description: (Last Updated: 04/28/2011)
Articles in Peer-reviewed Journals	Jamieson I, Sutton J, Strangman G. "System Concept for Autonomous Medical Diagnosis." Journal of the American Medical Informatics Association: Methods. [Ed. note: no publication information] , Jan-2004
Articles in Peer-reviewed Journals	Marshburn T, Strangman G, Strauss M, Sutton J. "SpaceDock: A performance task platform for spaceflight operations." Aviat Space Environ Med, submitted October 2003. , Oct-2003
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Presentation	Strangman, G "Near infrared brain imaging for space medicine " Jan-2003
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Presentation	Strangman, G., J. H. Thompson, M. Strauss, D. Boas, T. H. Marshburn and J. P. Sutton "Near Infrared Brain Imaging for Space Medicine " Jan-2004
Presentation	Thompson, J. H., K. Kwong, D. A. Boas, J. P. Sutton, and G. Strangman "Mayer-wave oscillations in the resting-state fMRI BOLD signal " May-2003